Jacobian-Dixmier Monoid

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Abstract

The explicit counterexample to the Jacobian conjecture in $n = 3$ [1, 2, 3] settles JC_n and DC_n for all $n \geq 3$, but for the present purposes it is only a seed. The object of study here is the structure it seeds: the two-sided action groupoid of $\mathrm{Aut}(W_n)$ on $\mathrm{End}(W_n)$, its π_0, its isotropy groups, and the monoid of bi-invariant subsets of $\mathrm{End}(W_n)$. We make the seed element $c_{\mathrm{bad}} \in \mathrm{End}(W_3) \smallsetminus \mathrm{Aut}(W_3)$ completely explicit via the Bass–Connell–Wright lift, record what it generates, and record the invariants presently able to separate classes. We then state a deliberately maximal conjecture: that the seed is a generator, and that nothing in the bi-invariant structure is invisible from it. The conjecture is written with the same unearned optimism on which the Jacobian conjecture itself survived for 87 years, and it is stated in order to be attacked. Section 7 specifies exactly what a disproof would consist of and how a candidate is certified, so that this document can serve simultaneously as the opening of a paper and as a self-contained problem specification for a future counterexample search, human or automated.

1 Jacobian and Dixmier Background

Throughout let k be a field of characteristic 0.

1.1 Definition ($JC_n(k)$) *Any polynomial endomorphism ϕ of Spec $k[x_1 \cdots x_n]$ over Spec k with Jacobian determinant 1 is actually a polynomial automorphism.*

1.2 Theorem (Normalization; Connell – van den Dries) *Let F be a polynomial endomorphism of $\mathbb{A}^n_k$ with constant Jacobian determinant $\lambda \in k^\times$.*

1. *For any $L \in \mathrm{GL}_n(k)$ the composite $L \circ F$ has constant Jacobian determinant $\det(L)\lambda$, and $L \circ F$ is an automorphism if and only if F is. Taking $L = \mathrm{diag}(\lambda^{-1}, 1, \ldots, 1)$, any counterexample with nonzero constant Jacobian determinant normalizes to one of determinant exactly 1, so $JC_n(k)$ as stated above is equivalent to the a priori stronger statement with determinant in $k^\times$.*
2. *[6] JC_n holds over every field of characteristic 0 if and only if it holds over $\mathbb{Q}$; in particular the truth value of $JC_n(k)$ does not depend on the choice of $k \supseteq \mathbb{Q}$.*

1.3 Definition ($W_n(k)$) *The Weyl algebra in n dimensions over k is the unital associative algebra with $2n$ generators q_i and ∂_i with commutation rules $[q_i, \partial_i] = -1$, all other pairs of generators commuting. This is the algebra of polynomial differential operators on Spec $k[q_1 \cdots q_n]$ as the variable names indicate.*

1.4 Definition ($\boldsymbol{DC_n(k)}$) *Any endomorphism of $W_n(k)$ is an automorphism. (Endomorphisms are automatically injective since $W_n(k)$ is simple; the content is surjectivity.)*

1.5 Theorem (Belov-Kanel, Kontsevich [7]) *$DC_n \implies JC_n$ and $JC_{2n} \implies DC_n$*

The former is the easy direction and it is all we need. The entirety of the aforementioned theorem is focused on the other half of the statement. More specifically we use the contrapositive of the easy half, in the strong constructive form of proposition 3.2 below, which produces a counterexample of DC_n from a counterexample of JC_n in the same dimension n, without passing through the stable $2n \leftrightarrow n$ correspondence.

2 The seed

The counterexample announced by Alpöge on 19 July 2026 [1, 2] is the map $F = (a, b, c) : \mathbb{A}^3 \to \mathbb{A}^3$ with

$$\begin{aligned} a &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ b &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ c &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

Its Jacobian determinant is the constant -2, and the map is generically 3-to-1 rather than injective. Both facts are finite symbolic computations. For the failure of injectivity it suffices to display a single collision; the fiber over the rational point $(a, b, c) = (0, 3, 0)$ consists of exactly the three points

$$\left(-\tfrac{2}{3},\ 3,\ 18\right), \qquad (0,\ 3,\ -36), \qquad \left(\tfrac{2}{3},\ -\tfrac{3}{2},\ \tfrac{45}{4}\right).$$

By Theorem 1.2(1), there is also the counterexample with determinant 1, namely

$$F_1 := \left(a,\ b,\ -\tfrac{c}{2}\right),$$

which has Jacobian determinant $\left(-\frac{1}{2}\right)(-2) = 1$ and the same fibers as F up to the linear change of target coordinates; the displayed fiber over $(0, 3, 0)$ is literally unchanged since c vanishes there. Write $d(F_1) := [k(x, y, z) : k(F_1)] = 3$ for the degree of the associated field extension; generically 3-to-1 is exactly the statement $d(F_1) = 3$.

One structural feature recorded in [2] will matter for the isotropy discussion below: a, b, c are homogeneous of weights $2, 1, -1$ for the grading $\deg(x) = -1$, $\deg(y) = 1$, $\deg(z) = 2$, so F (and F_1, whose normalization only rescales c) intertwines two one-parameter torus actions.

3 The lift

The seed enters the noncommutative world through the following construction, essentially the one underlying the easy direction of Belov-Kanel–Kontsevich and going back to Bass–Connell–Wright

[5]. Because the construction is purely first order, it has a classical output alongside the quantum one; the classical output carries the invariants we can currently compute, so we set both up at once.

3.1 Definition ($P_n(k)$) *Equip $P_n(k) := k[x_1, \dots, x_n, y_1, \dots, y_n]$ with the canonical symplectic Poisson bracket $\{x_i, y_j\} = \delta_{ij}$, $\{x_i, x_j\} = \{y_i, y_j\} = 0$. This is the semiclassical degeneration of $W_n(k)$, and it is Poisson-simple: a nonzero Poisson ideal is stable under all Hamiltonian derivations, hence under all partial derivatives, hence contains a nonzero constant. So Poisson endomorphisms of $P_n(k)$ are injective, exactly as endomorphisms of $W_n(k)$ are.*

3.2 Proposition (The lift) *Let $F = (F_1, F_2, F_3)$ be a polynomial endomorphism of $\mathbb{A}^3_k$ with* $\det JF = 1$*, and let $G := (JF^{\mathsf{T}})^{-1}$, a 3×3 matrix with polynomial entries since* $\det JF = 1$ *makes the inverse equal to the adjugate. Then:*

1. *$\Phi_F : x_i \mapsto F_i(x)$, $y_j \mapsto \sum_k G_{jk}(x)\, y_k$ defines a Poisson endomorphism of $P_3(k)$;*
2. *$\psi_F : q_i \mapsto F_i(q)$, $\partial_j \mapsto D_j := \sum_k G_{jk}(q)\, \partial_k$ defines an algebra endomorphism of $W_3(k)$;*
3. *if $d(F) > 1$ (in particular if F is generically 3-to-1), then neither Φ_F nor ψ_F is an automorphism.*

Proof (1) The brackets $\{F_i, F_j\} = 0$ are automatic since the F_i depend only on the x variables. For the mixed bracket, $\{F_i, \sum_k G_{jk} y_k\} = \sum_k G_{jk}\, \partial F_i / \partial x_k = (JF\, G^{\mathsf{T}})_{ij} = \delta_{ij}$ by the choice of G. The remaining bracket $\{\Phi_F(y_i), \Phi_F(y_j)\} = 0$ is the integrability identity

$$\sum_m \left(G_{jm} \frac{\partial G_{ik}}{\partial x_m} - G_{im} \frac{\partial G_{jk}}{\partial x_m} \right) = 0 \qquad \text{for all } i, j, k, \tag{3.3}$$

which holds because Φ_F is, pointwise, the cotangent lift of F: since $\det JF = 1$ everywhere, F is everywhere a local diffeomorphism, its cotangent lift $(x, p) \mapsto (F(x), (JF(x))^{-\mathsf{T}} p)$ is symplectic, and being symplectic is a pointwise condition needing no global inverse. For the specific F_1 of Section 2, equation (3.3) was additionally verified by direct symbolic computation; the entries of G are polynomials of total degree at most 11.

(2) In W_3 with the convention $[f(q), \partial_k] = -\partial f / \partial x_k$, the mixed commutators give $[\psi_F(q_i), \psi_F(\partial_j)] = -\delta_{ij}$ by the identical calculation, and expanding $[D_i, D_j]$ in normal form produces first-order operators whose coefficients are exactly the left side of equation (3.3): no ordering corrections arise because the $G_{jk}(q)$ commute among themselves and each D_j is of order one. So $[D_i, D_j] = 0$ and ψ_F is well defined.

(3) For Φ_F: the fraction field of the image is $k(F)(D_1, D_2, D_3)$ where $D_j = \sum G_{jk} y_k$; since G is invertible over $k(x)$, the y's lie in the $k(x)$-span of the D's, so $k(x, y) = k(x)(D_1, D_2, D_3)$ and the D's are algebraically independent over $k(x)$. Hence

$$[k(x, y) : \mathrm{Frac}(\operatorname{im} \Phi_F)] = [k(x)(D) : k(F)(D)] = [k(x) : k(F)] = d(F) > 1,$$

so Φ_F is not surjective.

For ψ_F: injectivity is free from simplicity of W_3. Suppose ψ_F were an automorphism. The centralizer of the commutative subalgebra $k[q] \subset W_3$ is $k[q]$ itself: writing an element in normal form $\sum_\alpha p_\alpha(q) \partial^\alpha$, commuting with each q_i kills every term with $\alpha \neq 0$. Automorphisms transport centralizers, so

$$\mathrm{Cent}_{W_3}\big(k[F]\big) = \mathrm{Cent}_{W_3}\big(\psi_F(k[q])\big) = \psi_F\big(\mathrm{Cent}_{W_3}(k[q])\big) = \psi_F(k[q]) = k[F].$$

But $k[q]$ is commutative and contains $k[F]$, so $k[q] \subseteq \mathrm{Cent}_{W_3}(k[F]) = k[F]$. Then each q_i is a polynomial in F_1, F_2, F_3, hence $k(x) = k(F)$ and $d(F) = 1$, contradicting $d(F) > 1$. $\square$

3.4 Corollary *$PC_3(k)$ (every Poisson endomorphism of $P_3(k)$ is an automorphism) and $DC_3(k)$ are both false, witnessed by*

$$c_{bad} := \psi_{F_1} \in \mathrm{End}(W_3(k)) \setminus \mathrm{Aut}(W_3(k)) \qquad \textit{and its symbol} \qquad \Phi_{F_1},$$

for every field k of characteristic 0.

This closes the counterexample chapter. Everything after this point treats c_{bad} purely as a distinguished element: the starting point, not the point.

4 The action groupoid and the bi-invariant monoid

$\mathrm{Aut}(W_n)$ acts by left and right multiplication on the monoid $\mathrm{End}(W_n)$

Because we have a set $\mathrm{End}(W_n)$ and a group $\mathrm{Aut}(W_n) \times \mathrm{Aut}(W_n)^{op}$ acting on it, we may form its action groupoid. Denote it $\mathrm{Aut}(W_n)\backslash\backslash \mathrm{End}(W_n)//\mathrm{Aut}(W_n)$ rather than putting ^{op} for the right action.

If we do π_0 in order to get the quotient instead, we get a pointed set G_n. However π_0 forgets two things: the multiplication law of $\mathrm{End}(W_n)$, and the isotropy. Both are part of the structure this paper is about.

4.1 Isotropy

The action groupoid is not equivalent to its π_0: each object ψ carries the isotropy group

$$\mathrm{Iso}(\psi) = \{(a_1, a_2) \in \mathrm{Aut}(W_n) \times \mathrm{Aut}(W_n)^{op} \text{ s.t. } a_1\psi a_2 = \psi\},$$

an invariant of the class strictly finer than the class itself. At the basepoint, $\mathrm{Iso}(\mathrm{id}) \cong \mathrm{Aut}(W_n)$ antidiagonally. The seed already has visible isotropy: the weighted homogeneity of (a, b, c) recorded in Section 2 means the torus automorphisms t_λ of W_3 with weights $(-1, 1, 2)$ on (q_1, q_2, q_3) and opposite weights on the ∂_i (these preserve the commutation relations, so are automorphisms) intertwine with the target torus of weights $(2, 1, -1)$ to fix ψ_{F_1} on the nose. So $\mathrm{Iso}(c_{\mathrm{bad}})$ contains a one-dimensional torus $\mathbb{G}_m(k)$, and computing it exactly is one of the concrete structure questions below (problem 7.2).

4.2 Semi-hypergroup

One way we could do the multiplication is sending $\big[a_1 f a_2\big] \times \big[a_3 g a_4\big] = \bigcup_{a \in \mathrm{Aut}(W_n)} \big[a_1 f a g a_4\big]$ where $a_i \in \mathrm{Aut}(W_n)$ and their appearance on the left or right does not affect the class but it does so in the middle between f and g. This is associative in the sense of hyperoperations because of the associativity of the starting monoids.

4.2.1 Linearization

A common way of dealing with this is to linearize over some field ℓ and then that union becomes a sum on the right hand side of a multiplication defined on basis elements. However that requires more control over the union than we have provided here in order to be useful.

4.3 Bi-invariant subsets

Let $\mathrm{BiInv}(\mathrm{End}(W_n))$ be the set of subsets of $\mathrm{End}(W_n)$ that are bi-invariant under the $\mathrm{Aut}(W_n)$ left and right actions. Under elementwise product $S \cdot T = \{st \text{ s.t. } s \in S, t \in T\}$ this is an honest monoid, not merely a hyperstructure: it works directly at the level of the power set rather than having only $P^*(X)$ on the right hand side of a hyperoperation as in all the flavors of hypergroup, and it also carries the complete lattice structure of unions, intersections and set differences of bi-invariant sets.

The empty set serves as an absorbing element on both sides. The subset $\mathrm{Aut}(W_n)$ serves as the left and right unit. The full set $\mathrm{End}(W_n)$ is also absorbing on the ideal of non-units in the following sense made precise by lemma 4.2.

4.1 Definition (Generated substructure) *For a family $\mathcal{S}$ of bi-invariant subsets, let $\langle \mathcal{S} \rangle$ denote the smallest subfamily of BiInv$(\mathrm{End}(W_n))$ containing $\mathcal{S}$, the unit $\mathrm{Aut}(W_n)$, and the absorbers $\emptyset$ and $\mathrm{End}(W_n)$, and closed under the monoid product, arbitrary unions, and set difference. (Intersections and complements follow from difference.)*

4.4 What the seed generates, and the invariants that see it

4.2 Lemma (No return to the unit) *Every element of every product class $[c_{bad}] \cdot [c_{bad}] \cdots [c_{bad}]$ is a non-surjective endomorphism. In particular no power of $[c_{bad}]$ meets the unit class $\mathrm{Aut}(W_n)$: non-surjectivity is a two-sided ideal condition on BiInv, and $[c_{bad}]$ generates a sub-semigroup avoiding the unit.*

Proof A typical element of an m-fold product is $a_0 \, \psi_{F_1} \, a_1 \, \psi_{F_1} \cdots a_{m-1} \, \psi_{F_1} \, a_m$ with $a_i \in \mathrm{Aut}(W_3)$; its image is contained in $a_0(\mathrm{im} \, \psi_{F_1}) \neq W_3$. □

4.3 Proposition (Degree grading on the classical side) *On the parallel structure built from* Aut *and* End *of the Poisson algebra $P_3(k)$, the assignment $[\Phi] \mapsto \deg[\Phi] := [k(x,y) : \mathrm{Frac}(im \, \Phi)]$ is well defined on classes and multiplicative on products: Poisson endomorphisms are injective by Poisson-simplicity, automorphisms have degree* 1*, and degrees multiply along composites. As* $\deg[\Phi_{F_1}] = 3$*, the classes of the successive powers of $[\Phi_{F_1}]$ have degrees $3, 9, 27, \ldots$ and are pairwise distinct: the classical monoid is infinite, graded by $(\mathbb{N}_{>0}, \times)$ on the classes reachable from the seed.*

On the quantum side the corresponding statements are open: the symbol of ψ_{F_1} along the order filtration is Φ_{F_1}, but automorphisms of W_n need not respect the filtration, so the degree does not descend to G_n for free. Producing the quantum degree is problem 7.1. This gap is itself informative: at present, *every* invariant we possess on G_n factors through classical shadows, and the conjecture below is exactly the assertion that nothing lives in the kernel of the shadows. That is the assertion one should distrust.

Note also that composites of lifts are again lifts up to direction of composition, $\psi_F \circ \psi_{F'} = \psi_{F' \circ F}$, so the m-th power class contains the lift of the iterate $F_1^{\circ m}$, which is generically 3^m-to-1.

5 The stabilization chain

5.1 Definition (f counterexample) *Let $f : [3] \to [n]$ be an injection of finite sets. For the counterexample $c_{bad} = \psi_{F_1}$ of DC_3 we have above, define the endomorphism $\mathrm{stab}_f(c_{bad})$ of the larger $W_n(k)$ by identity on q_i, ∂_i for those i not in the image and applying c_{bad} on the subalgebra $W_3(k)$ determined by f. This is well defined because the two subalgebras commute, and it is again non-surjective, so $DC_n(k)$ is false for all $n \geq 3$; on the commutative side this is the familiar stabilization by adjoining identity coordinates.*

By permutation automorphisms, when we quotient on the left and right by $\mathrm{Aut}(W_n)$ then we can ignore the f details but not until then.

5.2 Definition (Action groupoid chain) *Give shorthand $G_n \equiv \mathrm{Aut}(W_n)\backslash\backslash \mathrm{End}(W_n)//\mathrm{Aut}(W_n)$, then*

$$G_3 \longrightarrow G_4 \longrightarrow G_5 \longrightarrow \cdots$$

where the arrows are suppressing several arrows for all possible inclusions of $[n-1] \to [n]$. The choice of inclusion changes the morphism of action groupoids but it does not matter for the π_0 which is why the other arrows are suppressed in the notation.

5.3 Remark (Below the chain) The chain is drawn starting at G_3 because that is where non-triviality is proven. G_1 and G_2 remain conjecturally trivial, and their triviality is entangled with the surviving planar problem: triviality of G_2 is DC_2, which implies the still-open JC_2, while triviality of G_1 is DC_1, which would follow from JC_2 by [7]. Nothing below asserts anything about $n \leq 2$.$\diamondsuit$

6 The conjecture, stated to be attacked

6.1 Definition (Reachable elements) *Let $L_n \subseteq \mathrm{End}(W_n)$ be the set of all finite words*

$$a_0\, s_1\, a_1\, s_2 \cdots s_m\, a_m, \qquad m \geq 0, \quad a_i \in \mathrm{Aut}(W_n), \quad s_j \in \{\mathrm{stab}_f(c_{bad}) \text{ s.t. } f : [3] \hookrightarrow [n]\}.$$

Call an endomorphism, or its class in G_n, reachable *if it lies in L_n. The case $m = 0$ gives the unit class; by lemma 4.2 every reachable element with $m \geq 1$ is non-surjective.*

6.2 Conjecture (Remaining Optimistic Dixmier) *For every $n \geq 3$:*

1. *(orbit form) $\mathrm{End}(W_n) = L_n$. Every endomorphism of the Weyl algebra is a product of automorphisms and stabilized copies of the single seed c_{bad}; equivalently, the map from words to G_n is surjective.*
2. *(monoid form) Consequently $BiInv(\mathrm{End}(W_n)) = \langle\{[w] \text{ s.t. } w \in L_n\}\rangle$ in the sense of definition 4.1: every bi-invariant subset is built from reachable classes by products, unions, and set differences.*

Informally: the July 2026 map is not just a counterexample but a generator, and nothing in the bi-invariant structure of $\mathrm{End}(W_n)$ is invisible from it.

6.3 Remark (Epistemic status: this conjecture has no right to be true) The conjecture is stated at maximal strength, and the expectation here is that it is *false*. It is written in deliberate imitation of the epistemic position the Jacobian conjecture occupied for 87 years: no counterexample was found, the known obstructions to rigidity all vanished, and the absence of counterexamples was read as evidence for rigidity when it now appears to have measured only the difficulty of the search. That pattern — known obstructions vanish, rigidity is conjectured, rigidity fails once someone can actually search — is common in this exact neighborhood. The Markus–Yamabe conjecture held in the plane and fell to an explicit polynomial counterexample in dimension 3 [9]. The conjecture that all automorphisms of $k[x, y, z]$ are tame fell to the wildness of the Nagata automorphism [10]. JC_n itself fell in dimension 3 [1]. In each case the optimistic statement was defensible only because nothing was known to contradict it, which is precisely the situation of conjecture 6.2: by Section 4.4, every currently computable invariant of G_n factors through classical shadows, so the conjecture is exactly as strong as our current blindness. Its function is to be a sharp target. Following the strategy that produced the seed itself — pose the optimistic statement explicitly and search hard for the counterexample — a disproof would be at least as valuable as a proof, and Section 7 is written to make the target as easy to hit as possible. $\diamondsuit$

7 Disproof targets and certificates

This section is the specification. It is addressed to any future reader, human or automated, undertaking the search; it assumes only the definitions above, and it states what a counterexample to conjecture 6.2 is and how to certify one. It deliberately stops short of carrying out any of the attacks.

7.1 What a disproof consists of

A disproof of the orbit form is an endomorphism $\psi \in \mathrm{End}(W_n)$, $n \geq 3$, together with a proof that $\psi \notin L_n$. Since membership in L_n is not decidable by inspection, in practice a proof of non-membership means: a function ι on $\mathrm{End}(W_n)$, constant on bi-invariant classes, computable on both ψ and on all of L_n, with $\iota(\psi) \notin \iota(L_n)$. A disproof of the monoid form may instead exhibit a bi-invariant subset not in the generated family, or show the generated family is too small in some structural sense (see R3). The three routes below are ordered by expected accessibility.

7.2 Route R1: the degree invariant, through the classical window

By proposition 4.3, in the Poisson model $P_3(k)$ the degree $\deg[\Phi] \in \mathbb{N}_{>0}$ is bi-invariant and multiplicative, and every reachable class on the classical side has degree in $\{3^m \text{ s.t. } m \geq 0\}$, since each stabilized seed contributes a factor of exactly 3 and automorphisms contribute 1. Therefore:

Target. A Keller map F' on $\mathbb{A}^3$ (constant nonzero Jacobian, normalized to 1 by Theorem 1.2) with generic fiber degree $d(F') \notin \{1, 3, 9, 27, \ldots\}$ — for instance $d(F') \in \{2, 4, 5\}$ — disproves the Poisson analogue of the orbit form immediately, via $\Phi_{F'}$ and proposition 3.2. Announced follow-on constructions already claim families of counterexamples in $\mathbb{A}^3$ of every generic fiber degree ≥ 3 [4]; if any member of degree not a power of 3 is confirmed, this route closes on the classical side at once. To transport the conclusion to G_n itself and kill conjecture 6.2 as stated, combine with a solution to problem 7.1; alternatively, any direct proof that some bi-invariant quantum invariant refines d suffices.

Certificate. For a candidate F': (i) verify $\det JF' \in k^\times$ symbolically; (ii) compute $d(F') =$

$[k(x,y,z):k(F')]$ by Gröbner basis elimination, or by exhibiting a fiber with $d(F')$ points together with the cubic-style resultant relations bounding the degree above; (iii) record that $d(F')$ is not a power of 3.

7.3 Route R2: genuinely quantum classes

The reachable set L_n is manufactured entirely from one commutative seed through the first-order lift. Nothing known forces a general endomorphism of W_n to be of this shape.

Target. An endomorphism $\psi \in \mathrm{End}(W_n)$ that is not the lift of any polynomial map, and a bi-invariant invariant separating it from L_n. Candidate invariants to develop, in increasing ambition: the isomorphism type of W_n as a bimodule over $\operatorname{im}\psi$; growth data (Gelfand–Kirillov or Bernstein) of W_n relative to the image; the geometry of the branching locus of the symbol, e.g. the discriminant Δ and the loci p,q,r of [2] as bi-invariant data of a class rather than of a map; and reduction modulo p, where endomorphisms of $W_n(\overline{\mathbb{F}_p})$ act on the huge center and pick up characteristic-p symplectic invariants in the manner of [7, 8]. The last is the most promising machine for invariants that do *not* factor through the characteristic-zero classical shadow, which is exactly what remark 6.3 says we are missing.

7.4 Route R3: size

Target. Show G_n is too large to be reachable: for example, a family $\{\psi_t\}$ of endomorphisms over a positive-dimensional base together with an invariant as in R2 that is constant on classes, varies continuously in t, and takes values a single reachable class cannot absorb. Word length in L_n is countable data over each automorphism letter; a genuinely continuous modulus not accounted for by the automorphism letters would contradict the orbit form. This route requires no single explicit exotic ψ, only a well-behaved invariant, and may be the correct level of generality if R1 and R2 both stall.

7.5 Structure problems accompanying the search

7.1 Problem (Quantum degree) *Construct a bi-invariant, multiplicative* $\deg : G_n \to \mathbb{N}_{>0}$ *with* $\deg[\psi_F] = d(F)$ *for all lifts. Suggested sources: dimension of the skew field of fractions of* W_n *over that of the image; or the degree of the induced map on centers after reduction mod p, shown independent of* $p \gg 0$*. A solution makes proposition 4.3 quantum, proves the powers of* $[c_{bad}]$ *pairwise distinct in* G_n*, and upgrades any R1 counterexample to a full disproof of conjecture 6.2.*

7.2 Problem (Isotropy of the seed) *Compute* $\mathrm{Iso}(c_{bad})$ *exactly. Section 4.1 shows it contains* $\mathbb{G}_m$*; the question is whether the grading of [2] accounts for all of it. Isotropy groups are class invariants, so a mismatch of isotropy is itself a separation tool for R2.*

7.3 Problem (The product union) *Describe the hyperoperation at the seed: is* $[c_{bad}] \cdot [c_{bad}] = \bigcup_a [\psi_{F_1}\, a\, \psi_{F_1}]$ *a single class, finitely many, or infinitely many? Equivalently, how does the class of* $\psi_{F_1} a \psi_{F_1}$ *vary with the middle letter* a*? Any invariant solving problem 7.1 is constant on this union, so finer tools are needed; conversely, infinitude here would already show the linearization of Section 4 needs completed coefficients.*

7.4 Problem (Compatibility along the chain) *Determine whether the stabilization maps $G_n \to G_{n+1}$ are injective, and whether reachability is stable under de-stabilization: can an unreachable class in G_n become reachable in G_{n+1}? A negative answer localizes the search for R2 to $n = 3$.*

References

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